

Sri Sivasubramaniya Nadar College of Engineering

Department of Information Technology

Course Code: UIT2603	Course Name: INTERNET OF THINGS AND C PROGRAMMING
Date: 06-02-2026	Class Activity - 3
Class: VI IT B	

1. A software company wants to develop a Smart Task Workflow Manager to organize project tasks and their execution sequence. Each task contains task details and a link to the next task, allowing tasks to be created dynamically and connected in order, forming a workflow chain similar to a linked list. Design and implement a C program for this application. Your program must explicitly demonstrate the following C programming concepts: creation and use of structures, pointer usage within structures, implementation of self-referential structures, passing structures to functions (call by value), passing pointers to structures to functions (call by reference), and designing a function that returns a structure.
2. A small electronics store has asked you to create a Smart Inventory Analyzer that runs on a command line. The application must take command-line arguments for product information, classify goods using enum, optimize memory using union, enhance code readability using typedef, and set system restrictions using #define. There should be five logical sections to the application. Each component of your program should be identified and clearly mapped to the appropriate C concept - define, enum, typedef, union, and command-line arguments. Explain how each is utilized to optimize memory, achieve functionality, and make your code easier to read.